

Keycloak Setup Runbook

Project: Neuroforge Nexus (SDLC Platform)

1. Initial Realm Setup

The Realm isolates your application's users and configurations from the master server.

1. Log in to the Keycloak Admin Console (<http://localhost:8080/admin>).
2. In the top-left corner, click the dropdown menu (usually says "master") and click **Create Realm**.
3. Set the **Realm name** to:

neuroforge-nexus

4. Click **Create**. Ensure you are now operating inside this new realm.

2. Client Configuration (React App)

The Client represents your React frontend that will request authentication.

1. On the left navigation menu, click **Clients**.
2. Click **Create client**.
3. Set the **Client ID** to:

neuroforge-backend

4. Click **Next**, leave standard flow enabled, and click **Save**.
5. Scroll down to the **Access settings** section and configure the following exact values:

- **Valid redirect URIs:**

http://localhost:5173/*

- **Valid post logout redirect URIs:**

http://localhost:5173/*

- **Web origins:**

http://localhost:5173

4. Click **Save** at the bottom.

3. Enable User Registration

This setting allows your React app to route users to the Keycloak sign-up form.

1. On the left navigation menu, click **Realm settings**.
2. Go to the **Login** tab at the top.
3. Find the **User registration** toggle and switch it to **ON**.
4. Click **Save**.

4. User Profile & Custom Role Capture

By default, Keycloak doesn't ask for a user's role. We must enable the User Profile feature to capture it during registration.

1. Go to **Realm settings** > **User Profile** tab.
2. If disabled, click the toggle to enable **User Profile Enabled**.
3. Click **Add attribute**.

4. Set **Name** to:

user_role

5. Set **Display name** to:

Select Your Role

6. Under **Permissions**, check both **User** and **Admin** so the user can see and edit it.
7. Under **Required settings**, toggle **Required** to ON.
8. *(Optional but recommended)* Under **Validations**, add an option validator to restrict choices to specific SDLC roles (e.g., DEVELOPER, SCRUM_MASTER).
9. Click **Save**.

5. Token Mapping (JWT Configuration)

This final step ensures the custom role captured in Step 4 is actually placed inside the JWT so your Spring Boot backend can read it.

1. Go to **Clients** > select **neuroforge-backend**.

2. Click the **Client scopes** tab.

3. Click on the dedicated scope link named:

neuroforge-backend-dedicated

4. Click **Add mapper** > **By configuration**.

5. Select **User Attribute** from the list.

6. Fill in the exact following details:

- **Name:** role-mapper
- **User Attribute:** user_role (Must match the name from Step 4 exactly)
- **Token Claim Name:** app_role
- **Claim JSON Type:** String

7. Ensure **Add to ID token** and **Add to access token** are toggled **ON**.

8. Click **Save**.

Verification: Once these 5 steps are complete, you can safely test the flow. Clicking "Register" in React will open the Keycloak form, ask for the "Select Your Role" dropdown, and pass the selected role inside the JWT back to your application.